Jungseob **Lee**

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Summary

I do my research in a university lab, with far fewer GPUs than the companies that build the largest models. Working this way, I have seen how much a good answer from AI still depends on computing cost. I think a grandmother in a small town in Korea who asks about a symptom late at night should get the same good answer as an engineer in Silicon Valley, and get it just as fast. So I work on LLM efficiency, making models cheaper to train and faster to run, without making them worse.

Research Interests: Large Language Models, Natural Language Processing, Agentic LLM, LLM Efficiency, Efficient Reasoning

Education

Korea University *Seoul, Korea*
Integrated M.S./Ph.D. in Computer Science *2021 – Present*
NLP Lab, Advisor: Prof. Heuiseok Lim. Research focus: large language models, natural language processing, agentic LLM, LLM efficiency, and efficient reasoning. · **Expected: Feb. 2027**

Work Experience & Major Projects

VAETKI, Korean Sovereign AI Foundation Model *Jun 2025 – Jan 2026*
Post-training Project Manager, Korea University Representative *NC AI (NCSOFT)-led consortium*

- Served as post-training PM representing Korea University in Korea's national **Sovereign AI Foundation Model** program, organized by the Ministry of Science and ICT, an approximately KRW 200 billion national initiative with staged six-month evaluations running through 2027.
- The Korea University team (Prof. Heuiseok Lim) was selected as a core research partner in the NC AI (NCSOFT)-led grand consortium of 14 industry–academia–research institutions and 40 adopters (including POSCO DX, Lotte Innovate, KAIST, ETRI, and MBC).
- Led the full post-training pipeline for the VAETKI model, covering Supervised Fine-Tuning (SFT), Reinforcement Learning, and dataset curation.
- **VAETKI model:** an industry-specialized sovereign foundation model, open-sourced on Hugging Face on Dec 30, 2025. A 100B+ parameter Mixture-of-Experts architecture (∼11B active parameters at inference) with a custom Multi-head Latent Attention (MLA) reducing memory usage by up to ∼83%.
- Achieved state-of-the-art results on Korean language and culture (CLIcK), advanced reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [Model] [Technical Report]
- Press: [Korea University] [AI Times] [Newstomato]

Multilingual Post-Training Datasets, Sovereign AI Foundation Model Program *Aug 2026*
Dataset Lead, Korea University with NC AI *AI Hub, download for Korean nationals only*

- Led the construction and release of two post-training datasets of the program on AI Hub, Korea's national AI data platform, with NC AI (NCSOFT) as the prime contractor and Korea University building and extending the data.
- **High-Quality Multi-Turn Dataset**, multi-turn post-training dialogues in Korean, English, Japanese, and Chinese across six domains (general chat, code, general reasoning, math, culture, and long-context RAG), 664M tokens.
- **Synthetic & Instruction Dataset**, 858,960 model-synthesized instruction-response samples in the same four languages and six domains, with chain-of-thought reasoning, about 5.8 GB.
- [Multi-Turn Dataset] [Synthetic & Instruction Dataset]

KURE-v3, Korean-English Retrieval Embedding Model *2026*

Core Contributor, Korea University NLP & AI Lab and HIAI Institute *Data synthesis, training optimization, experiments*

- The newest dense model of the **KURE** family of Korean-English retrieval models (250+ GitHub stars), released on Oct 20, 2026 with open weights on Hugging Face and a public API.
- Synthesized the training data, 3M+ query-document triplets with mined hard negatives and false-negative filtering.
- Optimized training with teacher-score distillation (a min-max normalized KL loss), Matryoshka embeddings from 2,048 down to 256 dimensions, and inputs of up to 8,192 tokens.
- Ran the experiments on the nine MTEB(kor, v2) retrieval tasks and English RTEB against 20+ open retrievers.
- With 1.1B parameters KURE-v3 scores 0.803 average nDCG@10, above much larger dense retrievers such as Qwen3-Embedding-8B and the 27B harrier-oss-v1, and keeps 99% of its score at 1,024 dimensions.
- [Model] [GitHub] [API]

Cross-lingual Language Transfer for Large Language Models *2024 – 2025*

Lead Researcher (first author) *Published at ACL 2025 (Oral)*

- Proposed a cross-lingual optimization method that efficiently transfers English-centric LLMs to new target languages while preserving source-language ability, accepted as an **Oral** presentation at ACL 2025.
- [PDF]

Efficient Reasoning for Large Reasoning Models *2025 – Present*

Lead Researcher (first author) *Findings of EMNLP 2026, preprints under review*

- Leading a first-author line of work on training-free adaptive thinking budgets (DART) and stable efficiency training via adaptive correct-only rewards (ACOER) for large reasoning models. DART was accepted to **Findings of EMNLP 2026**.
- Co-first author of **ThinkFuse**, a trajectory-aware test-time fusion method for small reasoning models, accepted to **Findings of EMNLP 2026**.
- [DART] [ACOER] [Code]

Polyglot, Open-Source Multilingual Language Models *2022*

Open-Source Contributor, EleutherAI

- Worked on efficient language transfer of pretrained language models with cross-lingual post-training (XPT) and MAD-X adapters, and also took part in pretraining.
- [GitHub]

Awards & Honors

Outstanding Reviewer Award *2026*

The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego

Recognized by the ACL program committee for review quality and service to the conference. [Certificate]

Best Paper Award (Top 1) *2025*

Annual Conference on Human and Cognitive Language Technology (HCLT 2025)

For *KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*, by **J. Lee** (first author), M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, and H. Lim, from Korea University, Soongsil University, and Human-Inspired AI Research.

Minister of Science and ICT Award *2023*

ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)

Team of M. Kang, **J. Lee**, and S. Lee, for *KoCheckGPT*, Korean-specialized detection of LLM-generated documents for enterprise security, selected from 79 teams (263 participants) across 28 universities. [Press]

Publications

Underlined = me, * equal contribution. Full list on Google Scholar.

Top-Tier Conferences

The Hallucination Signal Is a Mean Shift: Why Simple Probes Suffice
J. Lee, J. Seo, H. Lim [arXiv] *EMNLP 2026 (Main)*

Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization
S. Lee, M. Kim, **J. Lee**, H. Lim [arXiv] [Code] *EMNLP 2026 (Main)*

DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models
J. Lee, S. Hong, S. Lee, J. Seo, J. Son, S. Eo, C. Park, H. Park, H. Moon, H. Lim [arXiv] [Code] *Findings of EMNLP 2026*

ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models
M. Kang*, **J. Lee***, J. Seo, H. Lim [arXiv] [Code] *Findings of EMNLP 2026*

Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [PDF] *ICLR 2026 (Poster)*

Cross-lingual Optimization for Language Transfer in Large Language Models
J. Lee, S. Hong, H. Moon, H.S. Lim [PDF] *ACL 2025 (Oral)*

Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation
J. Lee, H. Moon, S. Lee, C. Park, S. Eo, H. Ko, J. Seo, S. Lee, H.S. Lim [PDF] *Findings of ACL 2024*

PEEP-Talk: A Situational Dialogue-based Chatbot for English Education
S. Lee, Y. Jang, C. Park, **J. Lee**, J. Seo, H. Moon, S. Eo, S. Lee, B. Yahya, *et al.* [PDF] *ACL 2023 (Demo)*

International Journals

Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation
C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh [DOI] *Knowledge-Based Systems, 2026*

An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training
S. Son*, C. Park*, **J. Lee***, M. Shim, C. Lee, Y. Jang, J. Seo, J. Lim, H. Lim [DOI] *Expert Systems with Applications, 2025*

Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0
C. Park*, **J. Lee***, S. Son, K. Park, J. Jang, H. Lim [DOI] *Applied Sciences, 2024*

A Survey on Evaluation Metrics for Machine Translation
S. Lee*, **J. Lee***, H. Moon, C. Park, J. Seo, S. Eo, S. Koo, H. Lim [PDF] *Mathematics, 2023*

Doubts on the Reliability of Parallel Corpus Filtering
H. Moon, C. Park, S. Koo, **J. Lee**, S. Lee, J. Seo, S. Eo, Y. Jang, H. Kim, H. Lee, *et al.* [DOI] *Expert Systems with Applications, 2023*

K-NCT: Korean Neural Grammatical Error Correction Gold-standard Test Set Using Novel Error Type Classification Criteria
S. Koo, C. Park, J. Seo, S. Lee, H. Moon, **J. Lee**, H. Lim [PDF] *IEEE Access, 2022*

Other International Conferences

Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations
D.J. Lee*, **J. Lee***, S. Lee, S. Hong, S. Son, S. Eo, J. Seo, H. Lim [arXiv] [Code] *Findings of AACL-IJCNLP 2026*

Structured Clinical Memory for Safety-Consistent Diagnosis and Recommendation Generation in Long Medical Dialogues
D.J. Lee, **J. Lee**, E. Park, T. Whang, H. Lim *IEEE BIBM 2026 (Short Paper)*

QUAK: A Synthetic Quality Estimation Dataset for Korean-English Neural Machine Translation
S. Eo, C. Park, H. Moon, J. Seo, G. Kim, **J. Lee**, H.S. Lim [PDF] *COLING 2022*

Empirical Analysis of Noising Scheme based Synthetic Data Generation for Automatic Post-editing
H. Moon, C. Park, S. Lee, J. Seo, **J. Lee**, S. Eo, H.S. Lim [PDF] *LREC 2022*

Under Review

Predicting and Repairing Merge Collapse in Large Language Models
J. Lee, S. Lee, S. Eo, H. Moon, J. Seo, H. Lim [arXiv] [Code] *arXiv, Oct 2026*

Distilling Directional Verification
J. Lee, S. Eo, S. Hong, S. Lee, C. Park, J. Seo, H. Lim [arXiv] [Code] *arXiv, Oct 2026*

CAST: Cost-Aware Speculative Trees from One-Pass Block Drafters
J. Lee, S. Eo [arXiv] [Code] *arXiv, Oct 2026*

Refusal Localizes, the Damage Relocates: Safety Layers Under Few-Sample Fine-Tuning
J. Lee*, D.J. Lee*, S. Eo, S. Hong, S. Lee, H. Lim [arXiv] [Code] *arXiv, Oct 2026*

Recovering Off-Policy Supervision for Speculative Decoding
J. Lee, C. Park, S. Eo, H. Moon [arXiv] [Code] *arXiv, Sep 2026*

Risk-Controlled Selective LLM Answering by Pricing Label-Free Checks
D.J. Lee*, **J. Lee***, C. Park, H. Moon, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Where Activation Sparsity and KV-Cache Sparsity Cross in LLM Decoding
J. Lee, S. Lee, S. Hong, S. Eo, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Faster Block-Diffusion Serving with Distribution-Free Risk Guarantees
J. Lee, D.J. Lee, C. Park, S. Eo, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Vision Is Not Overhead: One-Pass Block Drafting for Lossless Speculative Decoding in Vision-Language Models
J. Lee, S. Hong, D.J. Lee, C. Park, J. Seo, S. Eo, H. Lim [arXiv] [Code] *arXiv, Sep 2026*

Answer-Conditioned Chains of Thought Degrade Verifiable-Reasoning Distillation in Large Language Models
J. Lee, S. Lee, S. Son, D.J. Lee, S. Han, S. Eo, H. Lim [arXiv] [Code] *arXiv, Jul 2026*

To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG
J. Lee, C. Park, H. Lim [arXiv] [Code] *arXiv, Jun 2026*

Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards
J. Lee*, S. Lee*, S. Hong, M. Kim, C. Park, H. Lim [arXiv] [Code] *arXiv, Jun 2026*

RenderRank: Learning to Rerank Text with Compressed Visual Tokens
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [arXiv] *arXiv, Sep 2026*

LAMAR: An Open Language-Aware Multilingual Alignment Reranker
S. Hong, Y. Jang, **J. Lee**, S. Lee, H. Lim [arXiv] *arXiv, Jul 2026*

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